AP BIOLOGY TOPIC 3.4

Photosynthesis: Light-Dependent Reactions, Photophosphorylation & Calvin Cycle

UNIT 3: CELLULAR ENERGETICS (12–16% EXAM WEIGHT)

ENE-1.I | ENE-1.J | PSII → PSI & RUBISCO

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 3.4 (PHOTOSYNTHESIS: LIGHT-DEPENDENT REACTIONS, PHOTOPHOSPHORYLATION & CALVIN CYCLE)

CED Core Vocabulary

Net Endergonic Redox Equation — 6 CO_2 + 6 H_2O + Light Energy → $C_6H_{12}O_6$ + 6 O_2 (ΔG = +686 kcal/mol) H_2O is oxidized (loses e^-/H^+) to form O_2 gas (proven by ^{18}O isotope tracing!).

Chloroplast Compartmentalization — Thylakoids & Grana (Stacks): Folded internal membranes containing chlorophyll a/b pigments, Photosystems II & I, ETC proteins (plastoquinone, cytochrome...

Stroma — Fluid-filled space outside the thylakoids (higher pH ~8); contains RuBisCO and enzymes for the Calvin Cycle .

4. Chemiosmosis (Photophosphorylation) — High [H^+] inside the thylakoid lumen flows down its electrochemical gradient back into the stroma through ATP Synthase , driving ADP + P_i → ATP .

Evolutionary Origin (Endosymbiosis) — Photosynthetic pathways evolved in prokaryotic cyanobacteria , which were engulfed by ancestral eukaryotes to become chloroplasts .

Thylakoid Lumen (Space) — Tiny enclosed interior volume where H^+ ions accumulate (low pH ~5) to rapidly build a steep proton-motive force.

2. Photolysis (Water Splitting) — An oxygen-evolving enzyme splits water inside the thylakoid lumen (2 H_2O → 4 H^+ + 4 e^- + O_2) to replace electrons lost from PSII .

Overview — Occurs in the stroma ; uses CO_2 + ATP + NADPH (from the light reactions) to synthesize the 3-carbon sugar G3P (glyceraldehyde-3-phosphate) , which is used to...

§2. PHOTOSYNTHESIS OVERVIEW & CHLOROPLAST ANATOMY

Light → Chemical Energy

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Photosynthesis Overview & Chloroplast Anatomy ➔

Net Endergonic Redox Equation ➔ **Evolutionary Origin (Endosymbiosis)** ➔ **Chloroplast Compartmentalization**

- **Net Endergonic Redox Equation:**
  6 CO_2 + 6 H_2O + Light Energy → $C_6H_{12}O_6$ + 6 O_2 (**ΔG = +686 kcal/mol**)
  - **H_2O is oxidized** (loses **e^-/H^+**) to form **O_2 gas** (proven by **^{18}O** isotope tracing!).
  - **CO_2 is reduced** (gains **e^-/H^+**) to synthesize **G3P / carbohydrates**.
- **Evolutionary Origin (Endosymbiosis):** Photosynthetic pathways evolved in prokaryotic **cyanobacteria**, which were engulfed by ancestral eukaryotes to become **chloroplasts**.
- **Chloroplast Compartmentalization:**
  - **Thylakoids & Grana (Stacks):** Folded internal membranes containing *chlorophyll a/b pigments, Photosystems II & I, ETC proteins (plastoquinone, cytochrome* ***b_6f*** *), and ATP synthase*. Site of the **Light-Dependent Reactions**.
  - **Thylakoid Lumen (Space):** Tiny enclosed interior volume where **H^+** ions accumulate (**low pH ~5**) to rapidly build a steep proton-motive force.
  - **Stroma:** Fluid-filled space outside the thylakoids (**higher pH ~8**); contains *RuBisCO* and enzymes for the **Calvin Cycle**.

§4. LINEAR (NONCYCLIC) VS. CYCLIC PHOTOPHOSPHORYLATION

Balancing ATP : NADPH

FEATURE	NONCYCLIC (LINEAR) FLOW	CYCLIC ELECTRON FLOW
Photosystems Used	Uses **BOTH PSII (P680) and PSI (P700)**	Uses **PSI (P700) ONLY**
Electron Path	**H_2O → PSII → ETC → PSI → $NADP^+$** (not recycled)	Excited **e^-** from **PSI cycle back through Cyt b_6f ETC** and return to PSI
Products Made	**ATP, NADPH, and O_2**	**ATP ONLY** (via proton pumping + chemiosmosis)
Not Produced	—	**NO NADPH** and **NO O_2** (no water splitting!)

[CORE AXIOM / REMEMBER] **Why Cyclic Flow Exists:** The Calvin Cycle consumes **more ATP (9) than NADPH (6)** per G3P made; cyclic photophosphorylation generates extra **ATP** to make up this stoichiometric deficit.

Support this site
Scan to donate

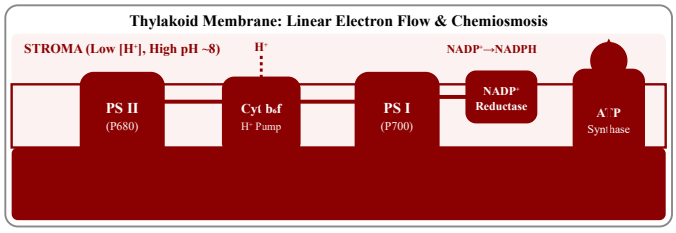
§3. LIGHT-DEPENDENT REACTIONS (THYLAKOID MEMBRANE)

PSII (P680) → PSI (P700)

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Light-Dependent Reactions (Thylakoid Membrane) ➔ **2. Photolysis (Water Splitting)** ➔ **4. Chemiosmosis (Photophosphorylation)**

- **1. Photoexcitation at Photosystem II (PSII — P680 acts FIRST):** Photons absorbed by antenna pigments excite electrons in reaction-center chlorophyll **P680** to a primary electron acceptor.
- **2. Photolysis (Water Splitting):** An oxygen-evolving enzyme splits water inside the thylakoid lumen ($2\ H_2O \rightarrow 4\ H^+ + 4\ e^- + O_2$) to **replace electrons lost from PSII**.
- **3. Linear Electron Transport Chain (ETC) & H^+ Pumping:** Excited electrons pass down from PSII through plastoquinone (**Pq**) → **Cytochrome Complex** → plastocyanin (**Pc**) to PSI. Energy released actively pumps **H^+ from the stroma into the thylakoid lumen**.
- **4. Chemiosmosis (Photophosphorylation):** High **[H^+]** inside the thylakoid lumen flows down its electrochemical gradient back into the stroma through **ATP Synthase**, driving $ADP + P_i \rightarrow ATP$.
- **5. Photosystem I (PSI — P700 acts SECOND) & NADPH Formation:** Light re-excites electrons at PSI; electrons pass via ferredoxin (**Fd**) to **$NADP^+$ Reductase**, reducing $NADP^+ + 2e^- + H^+ \rightarrow NADPH$ (the final electron acceptor!).



[COMMON AP EXAM FRQ TRAP] **High-Yield Exam Trap:** In photosynthesis, excited electrons end up in **NADPH** (**final electron acceptor is $NADP^+$**), NOT oxygen! Water (**H_2O**) is the initial electron donor for PSII.

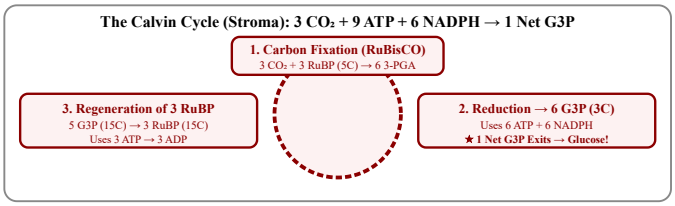
§5. THE CALVIN CYCLE: CARBON FIXATION IN THE STROMA

3 Phases → G3P Sugar

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

The Calvin Cycle ➔ **Overview** ➔ **Phase 1 — Carbon Fixation** ➔ **Phase 2 — Reduction** ➔ **Phase 3 — Regeneration of RuBP**

- **Overview:** Occurs in the **stroma**; uses **CO_2 + ATP + NADPH** (from the light reactions) to synthesize the 3-carbon sugar **G3P (glyceraldehyde-3-phosphate)**, which is used to build glucose, sucrose, starch, and cellulose.
- **Phase 1 — Carbon Fixation:** The enzyme **RuBisCO** catalyzes the attachment of inorganic **CO_2 (1C)** to the 5-carbon starting acceptor **RuBP (ribulose bisphosphate, 5C)** → forms an unstable 6C intermediate that immediately splits into two molecules of **3-PGA (3C)**.
- **Phase 2 — Reduction:** **ATP** phosphorylates intermediates and **NADPH** donates high-energy electrons (oxidized back to **$ADP + P_i$** and **$NADP^+$**, which return to the thylakoids!) to reduce 3-PGA into **G3P (3C)**. For every **3 CO_2** fixed, **6 G3P** are formed, and **1 net G3P exits** the cycle.
- **Phase 3 — Regeneration of RuBP:** The remaining **5 G3P** molecules (15 carbons total) are rearranged using additional **ATP** to regenerate **3 RuBP** (15 carbons) so the cycle can continue fixing **CO_2**.



[COMMON AP EXAM FRQ TRAP] **Classic FRQ Trap — What Happens When Light Is Turned Off?**

The Calvin Cycle requires light **indirectly**! Without light reactions running, **ATP and NADPH production stops immediately** → the Reduction & Regeneration phases halt → **3-PGA accumulates** while **G3P and RuBP levels drop** (or if **CO_2** is removed in light, **RuBP accumulates**!).

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 3.4 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 3.4: Photosynthesis: Light-Dependent Reactions, Photophosphorylation & Calvin Cycle ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 5, 6 & 13 (Enzyme Catalysis, Photosynthesis Leaf Disks & Respirometry): Enzyme assay (catalase/peroxidase spectrophotometry), Floating Leaf Disk Assay (ET_{50} = time for 50% of disks to float in $NaHCO_3$), and KOH Respirometry (KOH absorbs CO_2 gas as a solid precipitate so volume decrease directly measures O_2 consumption). Negative Control: boiled enzyme/boiled seeds or glass beads (corrects for ambient temp/pressure changes). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Reaction Rate = ΔProduct / ΔTime (initial linear slope V_0). Michaelis-Menten Kinetics: Competitive inhibitors increase apparent K_m without changing V_max; Noncompetitive/allosteric inhibitors decrease V_max without changing K_m. Gibbs Free Energy: $\Delta G < 0$ (exergonic/spontaneous, coupled via ATP → ADP + P_i, $\Delta G^{\circ\prime} = -7.3$ kcal/mol) vs. $\Delta G > 0$ (endergonic). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Bioenergetic Disruption Case Study: Cyanide (CN^-) and Carbon Monoxide (CO) bind irreversibly to Complex IV (Cytochrome c oxidase) of the mitochondrial ETC, blocking electron transfer to O_2, halting proton pumping, collapsing the H^+ gradient across the inner mitochondrial membrane, and arresting ATP Synthase oxidative phosphorylation. Conversely, 2,4-Dinitrophenol (DNP) and thermogenin (UCP1) make the inner membrane permeable to H^+, uncoupling electron transport from ATP synthesis and dissipating proton-motive force as heat.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 3.4 to prove why the claim holds.

Credits: mitali120501@gmail.com